MIL-H-62411(AT)
5 July 1985
SUPERSEDING
(see 6.3)

MILITARY SPECIFICATION

HEAT EXCHANGER, NBC SYSTEM GENERAL SPECIFICATION FOR

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, inspection, and acceptance requirements for the heat exchangers used in Nuclear Biological Chemical (NBC) system of the M1A1 tank and its three major components: heat exchanger, condenser and precooler.

1.2 Classification. Heat exchanger for the NBC system of the M1A1 tank shall consist of the following:

Heat Exchanger	12336771
Condenser	12336772
Precooler	12336768

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 4130

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam) Cleaner.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-H-62411(AT)/1	- Heat Exchanger.
MIL-C-62411(AT)/2	- Condenser.
MIL-P-62411(AT)/3	- Precooler.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-5624	- Turbine Fuel, Aviation, Grade JP-4 and JP-5.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, Preservation.
MIL-I-8500	- Interchangeability and Replaceability of Component Parts for Aircraft and Missiles.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-F-16884	- Fuel Oil, Diesel, Marine.
MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire-Resistant; Synthetic Hydrocarbon Base.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

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DRAWINGS
ARMY

12336768	- Precooler.
12336771	- Heat Exchanger.
12336772	- Condenser.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish heat exchangers which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent heat exchangers delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Detail specification. The requirements for the heat exchanger shall be as specified herein and in accordance with the applicable detail specifications for each of its major components. In the event of conflict between requirements of this specification and the referenced government documents, the requirements of this specification shall govern.

3.3 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free from defects which adversely affect performance or serviceability of the finished product (see 4.6.1).

3.3.1 Dissimilar metals. The use of dissimilar metals in intimate contact with each other shall be avoided unless suitably protected against electrolytic corrosion. Dissimilar metals are as specified in MIL-STD-889 (see 4.6.1).

3.4 Design and construction. The heat exchanger shall be constructed to the form and dimensions depicted on the drawings specified on the applicable specification sheet (see 4.6.2).

3.4.1 Weight. See applicable specification sheet.

3.4.2 Interchangeability. The heat exchanger shall be designed and fabricated in accordance with the interchangeability requirements of MIL-I-8500 (see 4.6.1).

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3.5 Performance.

3.5.1 Leakage.

3.5.1.1 External leakage. With the hot air passages (see 6.4) pneumatically pressurized as specified in applicable specification sheet, and with the heat exchanger immersed in water, there shall be no escaping bubbles that would indicate air leakage from the weld seams, pans, or side plates. Bubbles emerging from the cold air passages (see 6.4) are acceptable (see 4.6.4.1).

3.5.1.2 Internal leakage. With the hot air passages pressurized as specified at 73 ± 18 degrees Fahrenheit (°F) and isolated from the pressure source, the pressure loss in the first 20 seconds following isolation shall conform to the applicable specification sheet (see 4.6.4.2).

3.5.2 Proof pressure. The heat exchanger hot air passages shall withstand without damage or permanent deformation an air pressure as specified in the applicable specification sheet. The unit shall meet the requirements of 3.5.1 after subjection the the specified proof pressure (see 4.6.5).

3.5.3 Burst pressure. The heat exchanger hot air passages shall withstand without rupture an air pressure and temperature as specified in the applicable specification sheet (see 4.6.6).

3.5.4 Heat transfer. Heat transfer performance in terms of hot air effectiveness shall be not less than that shown in figure 1, in the applicable specification sheet (see 4.6.7).

3.5.5 Air pressure loss. Isothermal air pressure loss across the hot air passages shall not exceed the limit shown in figure 2, in the applicable specification sheet. The pressure drop presented in figure 2 is total-to-total, corrected to standard air density (see 4.6.8).

3.5.6 Environmental conditions.

3.5.6.1 Temperature.

3.5.6.1.1 High temperature. The heat exchanger shall meet the requirements of 3.5.1 during exposure to an operating temperature of 125°F. The unit shall also meet the requirements of 3.5.1 after exposure to a nonoperating storage temperature of 160°F (see 4.6.9.1).

3.5.6.1.2 Low temperature. The heat exchanger unit shall meet the requirements of 3.5.1 during exposure to an operating temperature of minus 60°F. The unit shall also meet the requirements of 3.5.1 after exposure to a nonoperating storage temperature of minus 60°F (see 4.6.9.1).

3.5.6.2 Humidity. The heat exchanger unit shall meet the requirements of 3.5.1 during exposure to a relative humidity up to 100 percent (see 4.6.9.2).

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3.5.6.3 Sand. The heat exchanger unit shall meet the requirements of 3.5.1 after exposure to sand particles of 0.01 to 1.00 millimeter (mm) diameter blown by air with a velocity of 3038 feet per minute (ft/min) (30 knots) against external component surfaces up to 1 hour. Sand concentration shall be 0.3 $\pm$ 0.2 grams per cubic foot (g/ft^3) (see 4.6.9.3).

3.5.6.4 Dust. The heat exchanger unit shall meet the requirements of 3.5.1 after exposure to dust laden air with dust particles of 0.0001 to 0.01 mm diameter and 6×10^{-9} grams per cubic centimeter (g/cm^3) blown against external component surfaces at an air velocity of 3544 ft/min (35 knots) (see 4.6.9.4).

3.5.6.5 Salt fog. The heat exchanger unit shall meet the requirements of 3.5.1 during exposure to a salt fog solution for a period of 48 hours. The salt fog solution shall be 5 percent by weight NaCl in 95 percent by weight of distilled water. The temperature in the exposure zone shall be maintained between 90 and 95°F. Fog density shall be approximately three quarts of salt solution per 10 cubic feet (ft^3) chamber volume per 24 hours (see 4.6.9.5).

3.5.6.6 Chemicals. The heat exchanger unit shall meet the requirements of 3.5.1 during exposure to vapors of and in direct contact with the following specifications for durations of up to 48 hours (see 4.6.9.6):

Fuel per VV-F-800, MIL-T-5624, MIL-G-3056 and MIL-F-16884

Hydraulic fluid per MIL-H-6083 and MIL-H-46170

Cleaning agents per P-C-437

3.5.6.7 Cleaning spray. The heat exchanger unit shall meet the requirements of 3.5.1 after exposure to a cleaning spray, either water jet or steam spray, using agents conforming to P-C-437 jet rinse immediately followed by a cold water rinse. The jet spray shall be applied perpendicularly to the surface being cleaned at a distance not less than 1.0 foot (ft) from the surface for steam cleaning and not less than 5.0 ft from the surface for water jet cleaning, and surface cleaned at a rate of 1.0 square foot per minute (ft^2/min). Nozzle pressure shall be a maximum of 110 pounds per square inch (psi). Nozzle orifice diameter shall be limited to a maximum of 1/4 inch for the water jet (see 4.6.9.7).

3.5.6.8 Fungus. The heat exchanger unit shall not support fungus growth when exposed to inoculation by spraying external surfaces with spore suspension (as defined by the "Specimen Inoculation" paragraph of MIL-F-13927) followed by exposure to ambient air temperatures of 80 to 84°F at relative humidities between 95 to 100 percent for a 28-day duration (see 4.6.9.8).

3.5.6.9 Vibration. The heat exchanger unit shall meet the requirements of 3.5.1 during exposure to sinusoidal vibration in accordance with table I in each of three mutually perpendicular axes (see figure 3). The vibration shall be imposed at a logarithmic sweep rate of 15 minutes per sweep cycle from 5 to 500 to 50 Hertz (Hz). The axes are defined in figure 4 (see 4.6.9.9).

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TABLE I. Vibration levels.

Axis	Frequency (Hz)	Amplitude
Vertical	5 to 25 25 to 57 57 to 500	± 1 gravity (g) 0.030 inch Double Amplitude ± 5 g
Longitudinal and Lateral	5 to 25 25 to 44 44 to 500	± 1 g 0.030 inch Double Amplitude ± 3 g

3.5.6.10 Basic shock. The heat exchanger unit shall meet the requirements of 3.5.1 during exposure to three shock impulses of 30 ± 3 gravity (g), 11.1 ± 1.1 milliseconds (ms) half sine wave applied in each direction of 3 mutually perpendicular axes (see figure 3 in the applicable specification sheet) (see 4.6.9.10).

3.5.6.11 Gun firing shock. The heat exchanger unit shall meet the requirements of 3.5.1 during exposure to shock impulses of 100 ± 1.0 g, 1.5 ± 0.2 ms applied in each direction of three mutually perpendicular axes (see figure 3 in the applicable specification sheet) (see 4.6.9.11).

3.5.6.12 Ballistic shock. The heat exchanger unit shall meet the requirements of 3.5.1 after subjection to ballistic shock impulses of 200 ± 20 g, 0.5 ± 0.1 ms applied in each direction of three mutually perpendicular axes (see figure 3 in the applicable specification sheet) (see 4.6.9.12).

3.6 Cross-reference to engineering and logistical support part number. Refer to table II for engineering and logistical support part number cross-reference interchangeability.

TABLE II. Cross-reference for part number interchangeability.

Engineering part number	Specification sheet part number
12336771 12336772 12336768	M62411/1 M62411/2 M62411/3

3.7 Marking. The heat exchanger shall be marked in accordance with MIL-STD-130 (see 4.6.2).

3.8 Workmanship. Workmanship shall be such as to provide a heat exchanger, including accessories, equipment, and loose parts, free from defects, such as: cracked, dented, loose, broken, or corroded parts/components (see 4.6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3.1).
   3. Control tests (see 4.5.3.2).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 $\pm$ 18°F
b. Barometric pressure 28.5 + 2.0 - 3.0 inches of mercury (Hg)
c. Relative humidity 50 $\pm$ 30 percent

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply heat exchangers that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of 3 unit and be inspected in the sequence listed in table III and as specified in table IV.

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4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select three from the first 200 heat exchanger units produced under the production contract for initial production inspection. Initial production units shall be numbered and inspected in the sequence listed in table III and as specified in table IV.

4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be prima facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government.

TABLE III. Sequence of tests.

Test sequence	Paragraph
High temperature	4.6.9.1
Low temperature	4.6.9.1
Humidity	4.6.9.2
Sand	4.6.9.3
Dust	4.6.9.4
Salt fog	4.6.9.5
Chemicals	4.6.9.6
Cleaning spray	4.6.9.7
Fungus	4.6.9.8
Vibration	4.6.9.9
Basic shock	4.6.9.10
Gun firing shock	4.6.9.11
Ballistic shock	4.6.9.12
Burst pressure	4.6.6

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TABLE IV. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Acceptance (100%)	Control
Materials and construction	3.3, 3.3.1, & 3.4.2	4.6.1	X			
Defects	3.4, 3.6, & 3.7	4.6.2	X	X		
Weight	3.4.1	4.6.3	X			
External leakage	3.5.1.1	4.6.4.1	X		X	
Internal leakage	3.5.1.2	4.6.4.2	X		X	
Proof pressure	3.5.2	4.6.5	X		X	
Burst pressure	3.5.3	4.6.6	X			
Heat transfer	3.5.4	4.6.7	X			
Air pressure loss	3.5.5	4.6.8	X			
High temperature	3.5.6.1.1	4.6.9.1	X			X
Low temperature	3.5.6.1.2	4.6.9.1	X			X
Humidity	3.5.6.2	4.6.9.2	X			
Sand	3.5.6.3	4.6.9.3	X			
Dust	3.5.6.4	4.6.9.4	X			
Salt fog	3.5.6.5	4.6.9.5	X			
Chemicals	3.5.6.6	4.6.9.6	X			
Cleaning spray	3.5.6.7	4.6.9.7	X			
Fungus	3.5.6.8	4.6.9.8	X			
Vibration	3.5.6.9	4.6.9.9	X			
First article		4.6.9.9.1	X			
Control test		4.6.9.9.2				X
Basic shock	3.5.6.10	4.6.9.10	X			
First article		4.6.9.10.1	X			
Control test		4.6.9.10.1				X
Gun firing shock	3.5.6.11	4.6.9.11	X			
First article		4.6.9.11.1	X			
Control test		4.6.9.11.2				X
Ballistic shock	3.5.6.12	4.6.9.12	X			

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the heat exchanger units of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

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4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective (see 6.2).

Classification	AQL
Major	2.5
Minor	4.0

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table V.

TABLE V. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 2.5% Defective	
101	Assembly, incomplete (see 3.4).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.4 thru 3.4.2).	SIE 1/
103	Identification marking, improper (see 3.7).	Visual
104	Faulty workmanship affecting performance (see 3.8).	Visual
Minor	AQL 4.0% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.4 thru 3.4.2).	SIE 1/
202	Faulty workmanship affecting appearance (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each heat exchanger unit shall be subjected to the acceptance tests specified in table IV.

4.5.3.2 Control tests. Control tests shall be conducted on one from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a six month period, nor less than one test in a 12 month period. The heat exchanger units shall be subjected to the tests specified in table IV.

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4.5.3.3 Failure. Failure of any heat exchanger unit to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.4 through 3.4.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.4 through 3.4.2, 3.6 and 3.7 shall be determined by examination for the defects listed in table V. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Weight. To determine conformance to 3.4.1, the heat exchanger shall be weighed.

4.6.4 Leakage.

4.6.4.1 External leakage. To determine conformance to 3.5.1.1, the heat exchanger unit shall be connected as shown in figure 4 in the applicable specification sheet. Immerse the unit in a water bath and apply a pneumatic pressure as specified in applicable specification sheet to the hot air passages. Examine the external weld seams, side plates, and pans. Verify that there are no escaping bubbles. Bubbles emerging from the fan air openings are acceptable during this test.

4.6.4.2 Internal leakage. To determine conformance to 3.5.1.2, with the same connection as in 4.6.4.1, a pneumatic pressure as specified in the applicable specification sheet shall be applied to the hot air passages. Close the two shut off valves in the supply line and open the vent valve to ensure complete isolation of the unit from the pressure source. The pressure shall be measured during the first 20 seconds following isolation.

4.6.5 Proof pressure. To determine conformance to 3.5.2, seal the air ports connecting the hot air passages to a pressure source, and leave the cold air ports open. The heat exchanger unit shall be placed in a safety enclosure as shown in figure 5 in the applicable specification sheet. Apply a pressure as specified in the applicable specification sheet to the hot air passages for no less than 5 minutes. At completion of the test verify that the unit has no visible damage or permanent deformation.

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4.6.6 Burst pressure. To determine conformance to 3.5.3, seal the air ports and leave the cooling air ports open. The heat exchanger shall be placed in a safety oven and a thermocouple shall be placed on the heat exchanger skin at a point of high mass as shown in figure 5 in applicable specifications sheet. Bring the oven temperature to the specified value in the applicable specification sheet until the unit temperature stabilizies within 5°F of the oven temperature, but not less than 1 hour. Apply a pressure as specified in the applicable specification sheet to the inlet air passages for not less than 5 minutes. Permanent deformation is permissible, however, the unit shall not burst as a result of the applied pressure.

4.6.7 Heat transfer. To determine conformance to 3.5.4, air flow shall be established in accordance with table VI of the applicable specification sheet, and hot air outlet temperature shall be measured for compliance with table V.

4.6.8 Air pressure loss. To determine conformance to 3.5.5, the isothermal pressure drop across the inlet air passages shall be measured at flows of 10, 14.7, and 20 pounds per minute [lb per min (± 5 percent)] specified in the applicable specification sheet at laboratory ambient temperature. The density correction factor shall be based on average air temperature and average air pressure in the air passages. Pressure loss shall not exceed the values of figure 2.

4.6.9 Environmental conditions.

4.6.9.1 Temperature. To determine conformance to 3.5.6.1, the assembly shall be tested as follows:

a. Operation.

1. Low temperature. The assembly shall be tested in accordance with MIL-STD-810, method 502.2, procedure II at -60°F.

2. High temperature. The assembly shall be tested in accordance with MIL-STD-810, method 501.2, procedure II at 125°F.

b. Storage.

1. Low temperature. The assembly shall be tested in accordance with MIL-STD-810, method 502.2, procedure I at minus 60°F for 24 hours.

2. High temperature. The assembly shall be tested in accordance with MIL-STD-810, method 501.2, procedure I at 160°F for 48 hours.

4.6.9.2 Humidity. To determine conformance to 3.5.6.2, place the heat exchanger unit in a humidity chamber and subject it to the humidity test specified in MIL-STD-810, method 507.2, procedure III. During humidity exposure, subject the heat exchanger unit to the performance test of 4.6.4.2. After exposure, the heat exchanger shall again pass the test of 4.6.4.

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4.6.9.3 Sand. To determine conformance to 3.5.6.3, the heat exchanger unit shall be tested in accordance with MIL-STD-810, method 510.2, procedure II with exceptions as noted in 3.5.6.3. Following exposure, the heat exchanger shall pass the test of 4.6.4.

4.6.9.4 Dust. To determine conformance to 3.5.6.4, the heat exchanger unit shall be tested in accordance with MIL-STD-810, method 501.2, procedure I with exceptions noted in 3.5.6.4. Following exposure, the heat exchanger shall pass the test of 4.6.4.

4.6.9.5 Salt fog. To determine conformance to 3.5.6.5, place the heat exchanger unit in a test chamber and subject it to the salt fog test specified in MIL-STD-810, method 509.2, procedure I. After the 48-hour exposure, conduct the test of 4.6.4.

4.6.9.6 Chemicals. To determine conformance to 3.5.6.6, the unit shall be operated while exposed to the vapors of and while in direct contact with the chemicals specified in 3.5.6.6 for a period of 48 hours. During exposure, the unit shall pass the test of 4.6.4.2. After exposure, the unit shall pass the test of 4.6.4.

4.6.9.7 Cleaning spray. To determine conformance to 3.5.6.7, the heat exchanger unit shall be subjected to a cleaning spray test as specified in 3.5.6.7. After exposure to the cleaning spray, the heat exchanger unit shall be visually examined externally for any signs of deterioration and shall pass the test of 4.6.4.

4.6.9.8 Fungus. To determine conformance to 3.5.6.8, the heat exchanger shall be tested in accordance with MIL-STD-810, method 508.3, except the condition of 3.5.6.8 shall apply.

4.6.9.9 Vibration. To determine conformance to 3.5.6.9, the heat exchanger shall be mounted using the installed interface configuration for its intended application or equivalent. The heat exchanger shall be tested in accordance with MIL-STD-810, method 514.3, category 8, procedure I, the vibration shall be imposed at a logarithmic sweep rate of 15 minutes per sweep while cycling between 5 to 500 to 5 Hz. Conduct the test of 4.6.4.2 during each axis. After vibrating the unit shall pass the test of 4.6.4.

4.6.9.9.1 First article. For first article testing the heat exchanger shall be subjected to a sinusoidal vibration test in accordance with the requirements of table I for a total of 60 minutes in each of three orthogonal axes.

4.6.9.9.2 Control tests. For control testing the exposure time shall be 90 minutes.

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4.6.9.10 Basic shock. To determine conformance to 3.5.6.10, the heat exchanger shall be mounted using the installed interface configuration for its intended application or equivalent. The heat exchanger shall be tested in accorance with MIL-STD-810, method 516.3, procedure I. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes for total of 18 shocks. At the conclusion of this test, the heat exchanger for shall evidence no damage and shall pass the test of 4.6.4.

4.6.9.10.1 First article. For first article testing, the shock amplitude shall be 30 ± 3 g, 11 ± 1.1 ms.

4.6.9.10.2 Control tests. For control testing, the shock pulse shall be 25 ± 2.5 g, 11 ± 1.1 ms.

4.6.9.11 Gun firing shock. To determine conformance to 3.5.6.11, the heat exchanger shall be mounted using the installed interface configuration for its intended application or equivalent, the heat exchanger shall be tested in accordance with MIL-STD-810, method 516.3, procedure VII. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes for a total of 18 shocks. At the conclusion of this test, the heat exchanger shall evidence no damage and shall pass the test of 4.6.4.

4.6.9.11.1 First article. For first article testing peak amplitude shall be 100 ± 10 g, 1.5 ± 0.2 ms.

4.6.9.11.2 Control tests. For control testing the peak amplitude shall be 75 ± 8 g, 1.5 ± 0.2 ms.

4.6.9.12 Ballistic shock. To determine conformance to 3.5.6.12, the heat exchanger shall be mounted using the installed interface configuration for its intended application or equivalent, the heat exchanger shall be tested in accordance with MIL-STD-810, method 516.3, procedure VII. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes for a total of 18 shocks. Peak amplitude shall be in accordance with 3.5.6.11. At the conclusion of this test, the heat exchanger shall evidence no damage and shall pass the test of 4.6.4.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The three heat transfer components covered by this specification serve to cool and dry M1 Tank engine bleed air for processing through the NBC system. Although the three units have different names, they are referred to herein by the generic term heat exchanger.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and ate of the applicable detailed specifications.
c. Number and date of the applicable drawing (see 3.4).
d. If first article samples are not required (see 3.1).
e. If responsibility for inspection shall be other than as specified (see 4.1).
f. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
g. If inspection conditions shall be other than as specified (see 4.3).
h. If preproduction inspection is required (see 4.4.1).
i. If initial production inspection is not required (see 4.4.2).
j. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Supersession data. This military specification supersedes the General Dynamics documents listed below:

SC-12336771, dated 29 March 1984
SC-12336772, dated 29 March 1984
SC-12336768, dated 29 March 1984

6.4 Definitions. Air flow terms used in this specification are defined as follows:

a. Hot air. Engine bleed air to be cooled in the heat exchanger.
b. Cold air. Air used as the cooling medium in the heat exchanger.

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